Practice questions 28 October 2026

Use $\pi = \dfrac{22}{7}$. For every question without a picture, first draw a rough diagram and mark all the given measurements on it. Show all your steps.

1. A circular lotus pond in a temple garden is fenced once with 132 m of wire. Draw a rough diagram.

(a) Find the radius of the pond.

(b) A second fence is put up 7 m outside the first fence, all the way round. Find how much more wire the second fence needs than the first.

(c) Another pond has a radius of 35 m. It also gets a second fence 7 m outside the first one. Will the extra wire needed be more than, less than, or the same as your answer to (b)? Explain.

2. Say whether each statement is true or false. Give a reason for each answer.

(a) The value of π is exactly $\dfrac{22}{7}$.

(b) For every circle, the circumference divided by the diameter gives the same number.

(c) Arjun measures a bangle. Its circumference is 22 cm and its diameter is 7 cm. Arjun says, "This proves that π is exactly $\dfrac{22}{7}$."

3. A railway line goes round a curve. The curve is an arc of a circle of radius 210 m.

(a) A train goes round the curve and turns through an angle of $60°$. Draw a rough diagram. Find the distance the train travels on the curve.

(b) On another curve of the same radius, a train travels 330 m. Find the angle through which the train turns.

4. In a 200 m race on a school track, each runner runs one bend and one straight. The bend in every lane is a semicircle. Each lane is 1.4 m wide. All runners finish at the same line. Assume that each runner runs along the inner edge of the lane. Draw a rough diagram of lanes 1 and 2.

(a) Explain why the runner in lane 2 must start ahead of the runner in lane 1.

(b) Find how far ahead of the lane 1 runner the lane 2 runner must start.

(c) Find how far ahead of the lane 1 runner the lane 4 runner must start.

(d) Ravi says, "If the track had a bigger bend, the answer to (b) would change." Is Ravi right? Explain.

5. A garden bed ABCD is shaped like a parallelogram. AB = 18 m and AD = 12 m. The perpendicular distance between the sides AB and DC is 10 m.

(a) Draw a rough diagram. Mark all the lengths and show the 10 m distance with a dotted line.

(b) Raju finds the area like this: Area = $18 \times 12 = 216$ m^2. Explain Raju's mistake. Find the correct area of the bed.

(c) Find the perpendicular distance between the sides AD and BC.

6. A strip of cloth has two straight parallel edges that are 7 cm apart. Kavi draws two triangles, PQR and SQR, on the strip. Both triangles have the side $QR = 12$ cm along the lower edge. The points P and S are on the upper edge. In triangle PQR, the angles at Q and R are acute. In triangle SQR, the angle at Q is obtuse.

(a) Draw both triangles in one rough diagram. Show the height of each triangle with a dotted line.

(b) Find the area of each triangle.

(c) Priya says, "Triangle SQR has the larger area, because its sides are longer." Is Priya right? Explain.

7. A glass panel $ABCD$ in a lamp is shaped like a parallelogram, and its area is 96 cm^2. Two thin metal strips run along the diagonals AC and BD. They meet at O and divide the glass into four triangular pieces. Draw a rough diagram.

(a) Explain why triangles AOB and BOC have equal areas. (Hint: what is BO in triangle ABC?)

(b) Explain why all four pieces have equal areas. Find the area of each piece.

(c) Neha says, "The four pieces are congruent." Is Neha right? Explain.

8. A triangular plot ABC has sides $BC = 90$ m, $CA = 100$ m and $AB = 170$ m.

(a) Find the area of the plot using Heron's formula.

(b) Use your answer to (a) to find the shortest distance from the corner A to the line BC.

(c) Draw a rough diagram showing this shortest distance AD. Does D lie on the side BC, or on BC extended beyond C? Find CD and check your answer to (b) using AB.

9. A garden bed is shaped like an isosceles triangle. Its perimeter is 64 m and its unequal side (the base) is 24 m. Draw a rough diagram.

(a) Find the length of each equal side.

(b) Find the area of the bed using Heron's formula.

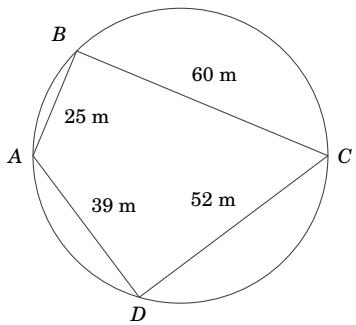
(c) Find the area again in a different way, using the height on the base. Check that both answers agree.

10. A plot of land $ABCD$ has all four corners on a circle, as shown. $AB = 25$ m, $BC = 60$ m, $CD = 52$ m and $DA = 39$ m.

(a) Find the area of the plot using Brahmagupta's formula.

(b) The diagonal AC is a diameter of the circle, and $AC = 65$ m. So the angles at B and D are right angles. Find the area of the plot again by splitting it along AC into two triangles. Check that it agrees with (a).

(c) A rectangle 12 m by 5 m also has all four corners on a circle. Use Brahmagupta's formula to find its area. Does the formula give the answer you expect? Explain.



11. Long ago, Baudhāyana showed how to change a rectangle into a square of the same area. Follow his steps for a rectangular field 18 m long and 8 m wide. Draw a diagram for each step.

(a) Cut off a square of side 8 m from one end of the rectangle. What are the length and breadth of the strip that is left?

(b) Cut the strip into two equal strips, each 8 m long. Place one strip along the right side of the square and the other along the top of the square. This makes an L-shape. Show that the L-shape fits inside a square of side 13 m, with a small square missing at one corner. Find the side of the missing square.

(c) Find the area of the L-shape in two ways: from the rectangle, and as (big square) − (small square).

(d) Baudhāyana now draws a right triangle with hypotenuse 13 m and one side 5 m. Find the third side. Explain why a square on this third side has the same area as the field.

12. The top of a round table has a circumference of 220 cm.

(a) Find the radius of the table top.

(b) Find the area of the table top.

(c) A bigger round table has a radius twice as long. Ravi says, "Its top has twice the area." Is Ravi right? Find the area of the bigger table top and explain.

13. A car's wiper arm is 21 cm long and turns about one end, O. The rubber blade is fixed to the outer part of the arm, from 7 cm to 21 cm from O. The arm turns through an angle of 90°.

(a) Draw a rough diagram. Shade the part of the glass that the blade wipes.

(b) Find the area of the glass that the blade wipes.

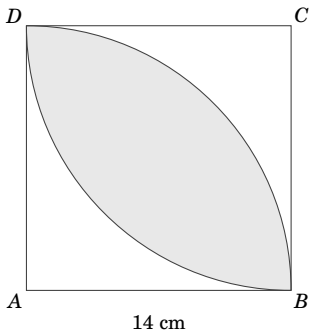
(c) Find the length of the path of the tip of the arm.

14. A floor tile $ABCD$ is a square of side 14 cm. Two quarter circles are drawn inside it: one with centre A and radius AB, and one with centre C and radius CB. Both arcs go from B to D. The leaf-shaped part between the two arcs is shaded.

(a) Find the area of the shaded leaf.

(b) Find the perimeter of the shaded leaf.

(c) Meena finds the area of the leaf like this: (area of one quarter circle) + (area of the other quarter circle) − (area of the square). Explain why her method works, and check that it gives the same answer.



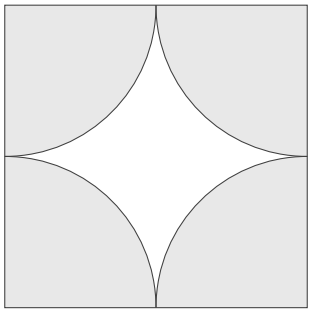
15. A square garden has side 28 m. At each corner there is a flower bed shaped like a quarter circle of radius 14 m, with its centre at the corner of the garden. The rest of the garden is lawn.

(a) Find the area of the lawn.

(b) Find the perimeter of the lawn.

(c) The gardener makes the flower beds smaller, with radius 7 m. Find the new area of the lawn and the new perimeter of the lawn.

(d) Did the perimeter of the lawn increase or decrease in (c)? Explain why.



28 m